

Board Name

3D rendering

DESIGN NOTES:
Example text for
informational
design notes.

DESIGN NOTES:
Example text for
cautionary design
notes.

DESIGN NOTES:
Example text for
critical design
notes.

LAYOUT NOTES:
Example text for
critical layout
guidelines.

DEBUG NOTES:
Example text for
debug notes.



Flight Computer

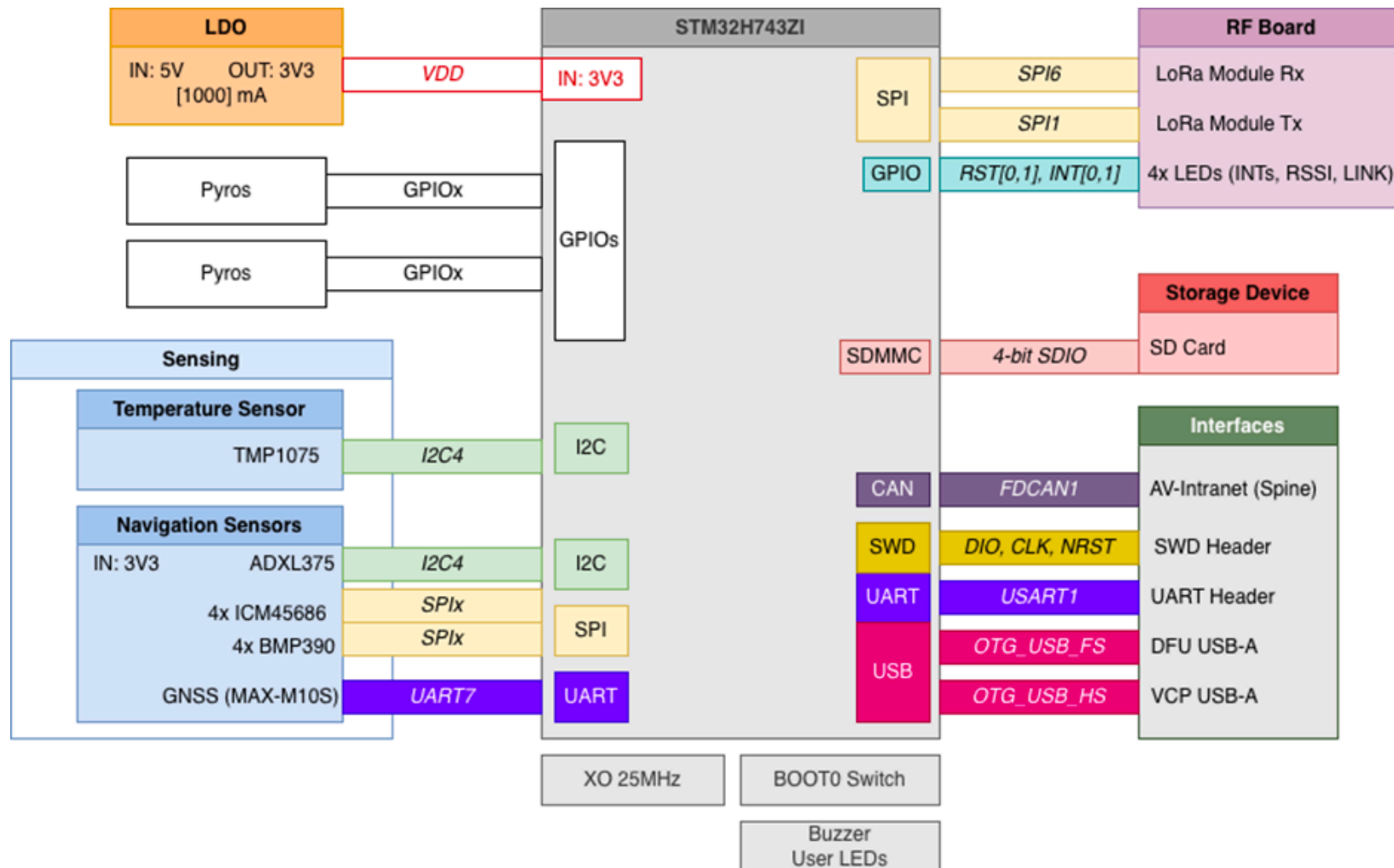
Author: Max Wei Ciriani
Date: 2025-11-23 ©EPFL Rocket Team 2026

Sheet: Root
Size: A4
Software: KiCad E.D.A. 9.0.6

Rev:
Id: 1/16



Block Diagram



ZENITH-FC

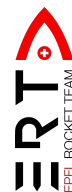
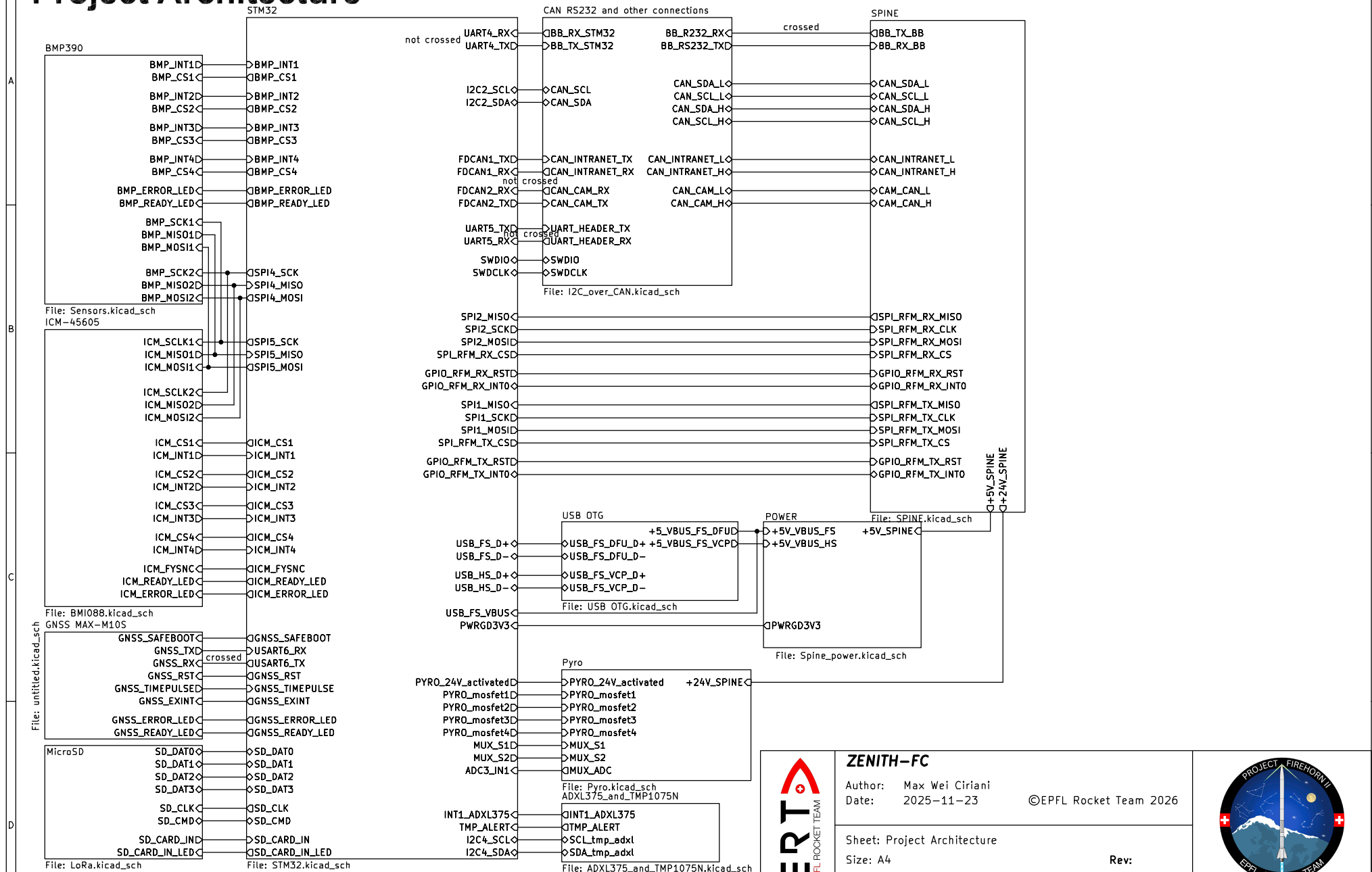
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Project Architecture



ZENITH-FC

Author: Max Wei Ciriani

Date: 2025-11-23

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Sheet: Project Architecture

Size: A4

Software: KiCad E.D.A. 9.0.6

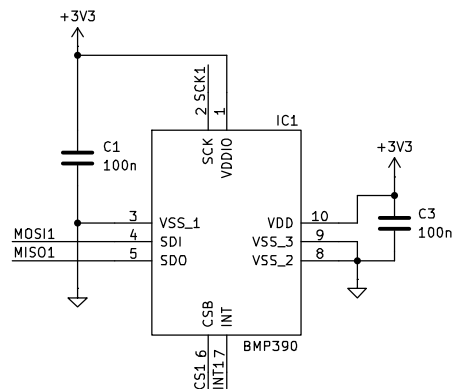
Rev:

Id: 3/16

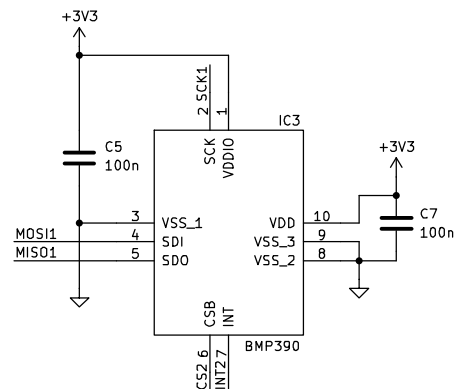


BMP390

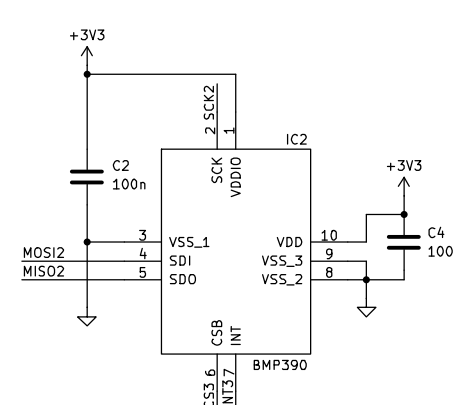
BMP390 #1



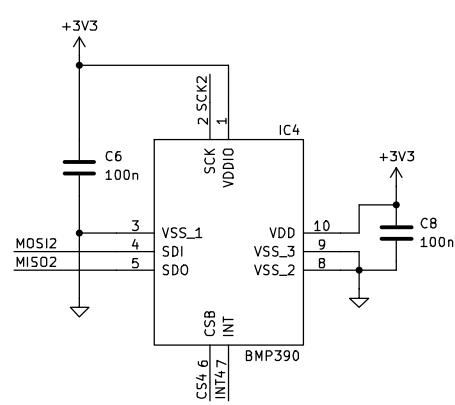
BMP390 #2



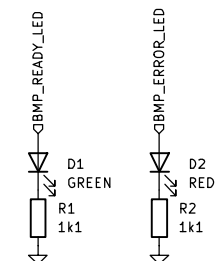
BMP390 #3



BMP390 #4



Labels	
MOSI1	↔BMP_MOSI1
MISO1	↔BMP_MISO1
SCK1	↔BMP_SCK1
MOSI2	↔BMP_MOSI2
MISO2	↔BMP_MISO2
SCK2	↔BMP_SCK2
CS1	↔BMP_CS1
INT1	↔BMP_INT1
CS2	↔BMP_CS2
INT2	↔BMP_INT2
CS3	↔BMP_CS3
INT3	↔BMP_INT3
CS4	↔BMP_CS4
INT4	↔BMP_INT4



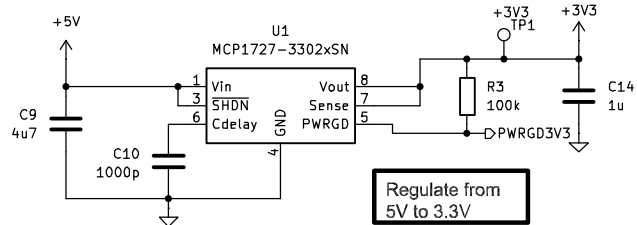
DESIGN NOTES:
In case of SPI malfunction, the 4 sensors are divided in two buses, sensors 1,2 have one bus, and sensors 3,4 have another bus.

DESIGN NOTES:
The datasheet mentions SDI and SDO, the equivalents are:
SDI = MOSI
SDO = MISO

SENSORS

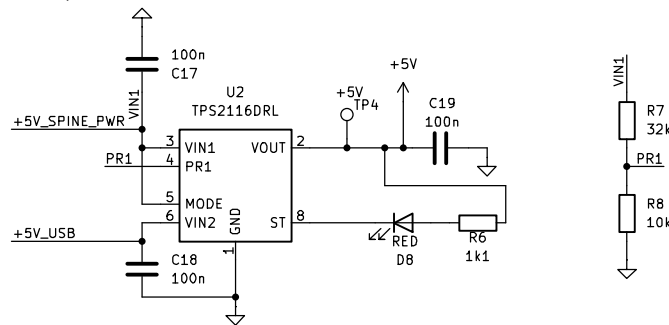
	BMP390 Author: Max Wei Ciriani Date: 2025-11-23 ©EPFL Rocket Team 2026		
	Sheet: BMP390 Size: A4 Software: KiCad E.D.A. 9.0.6		
	Rev: 4/16		

5V->3V3 LDO



The SHDN input is used to turn the LDO output voltage on and off. When the SHDN input is at a logic-high level, the LDO output voltage is enabled. When the SHDN input is pulled to a logic-low level, the LDO output voltage is disabled. When the SHDN input is pulled low, the PWRGD output also goes low and the LDO enters a low quiescent current shutdown state where the typical quiescent current is 0.1 μ A.

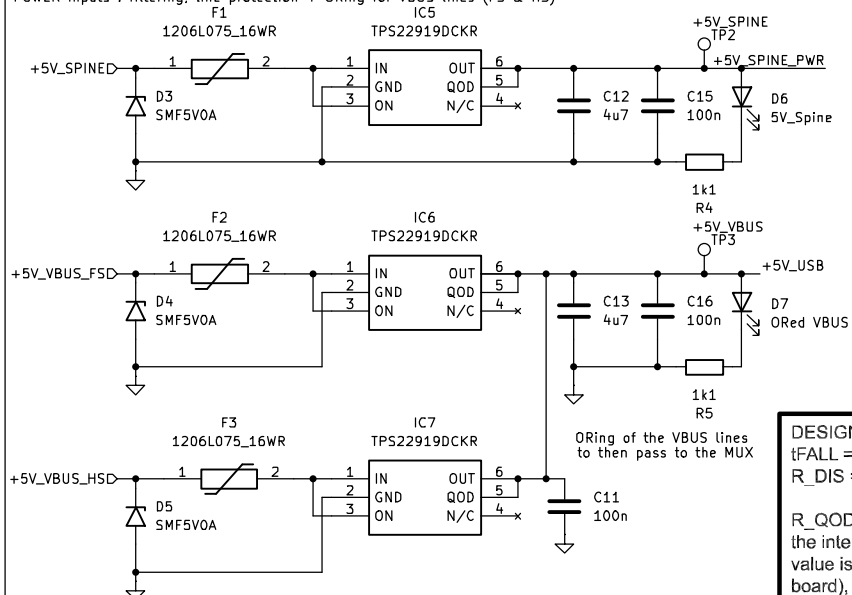
Power MUX : Priority to SPINE, then VBUS



DESIGN NOTES:
VIN1 not being used : LED ON

DESIGN NOTES:
Vref = 1V (0.92-1.08)
If PR1 > Vref -> Vin1
If PR1 < Vref -> Vin2
So VinCritical = (R1+R2)/(R2)*Vref = 4.2V

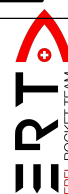
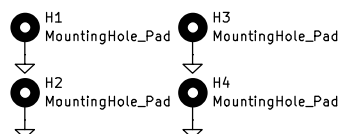
POWER inputs : filtering, line protection + ORing for VBUS lines (FS & HS)



DESIGN NOTES:
Output current up to 1.5A
safely
 $V_{drop} = I_{max} \times R = 500mA \times 125m\Omega = 62.5mV \rightarrow$
basically an ideal diode

DESIGN NOTES:
 $t_{FALL} = 2.2 \times (R_{DIS} || R_L) \times C_L$ where
 $R_{DIS} = R_{QOD} + R_{PD}$
 R_{QOD} is 0 (cause shorted) and R_{PD} is the internal resistor. Shorted, cause C_L value is pretty high (all of the capas on the board), so I don't want a larger fall time.

SPINE & POWER



Power Management

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Date: 2025-11-23

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Sheet: POWER

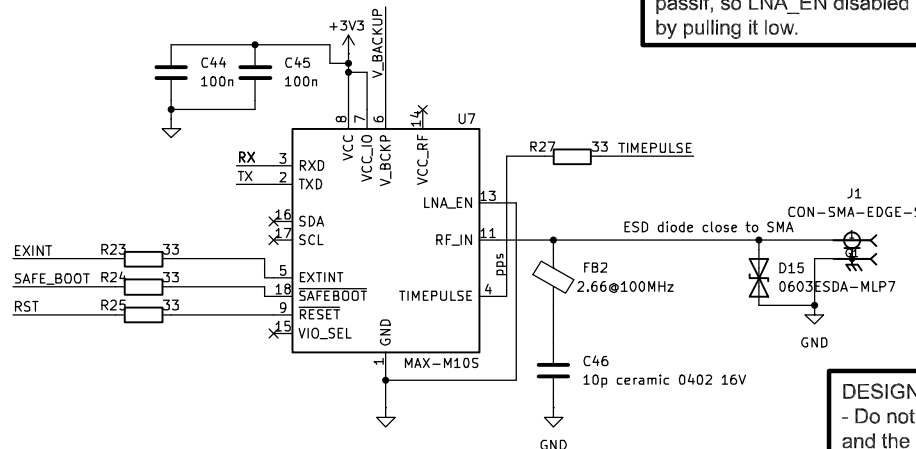
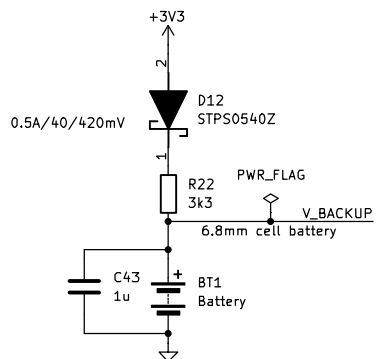
Size: A4

Software: KiCad E.D.A. 9.0.6

Rev:

Id: 5/16





DESIGN NOTES:
Ceramic patch antennas are passif, so LNA_EN disabled by pulling it low.

DESIGN NOTES:
LNA_EN pulled low to disable the optional external LNA (low noise amp)

DESIGN NOTES:
The RF line must match 50 ohms of impedance.

DESIGN NOTES:
VIO_SEL line must be left openned if powered by 3v3.

DESIGN NOTES:
To enter safeboot mode, set pin to low at receivers startup, otherwise, leave it open

DESIGN NOTES:
- Do not route anything other than the ground plane and the RF line! No signal trace, no power traces, no vias.
- ESD diode close to SMA connector

DESIGN NOTES:
The TIMEPULSE pin outputs a clock (default 1Hz, can be changed using the CFG-TP-* config. group (see datasheet page 48 (section 3.9 time pulse))). However, on startup it also shares the same internal IC as SAFE_BOOT_N. If the pin is low at startup, the receiver will enter safeboot mode, so make sure that there is no load that could pull the pin low at startup.

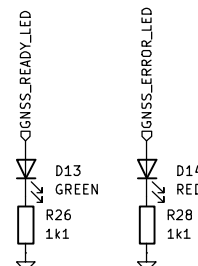
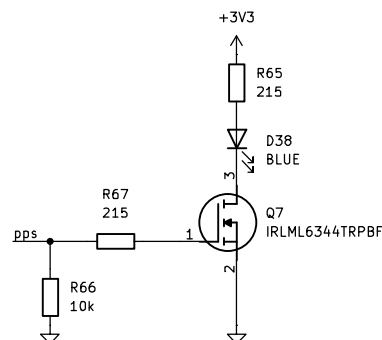
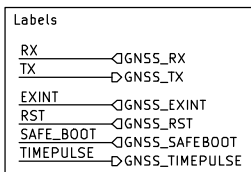
DESIGN NOTES:
The critical length for the microstrip (outer trace) is around 8.6mm. So above 8.6mm, you need to consider RF effects.

DESIGN NOTES:
Place de 33Ohmes resistors close to the MCU

DESIGN NOTES:
2.1.2 Internal LNA mode configuration
u-blox 10 receivers support three modes for the internal low-noise amplifier (LNA): normal gain, low gain, and bypass mode. The MAX-M10S default is the low-gain mode. With a high-gain external active antenna, the bypass mode can be used. The normal-gain mode is not recommended for MAXM10S.

The internal LNA mode can be configured at run time in BBR and RAM memory using the configuration item CFG-HW-RF_LNA_MODE and applying a software reset by sending UBX-CFGRST message. Refer to Forcing receiver reset for more information.

LAYOUT NOTES:
Baud rate [bt/s]:
Min = 9600
Max = 921600



GNSS MAX-M10S

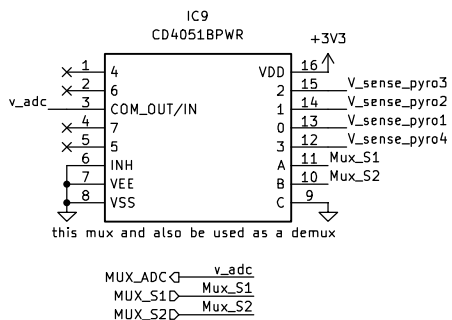
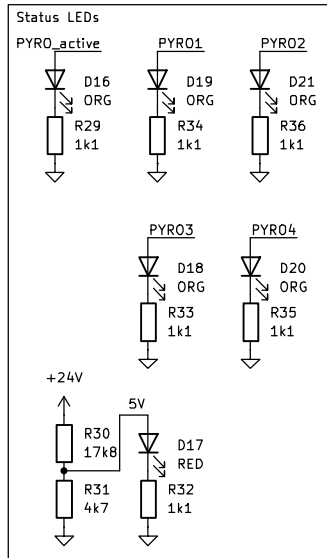
Author: Max Wei Ciriani
Date: 2025-11-23

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Sheet: GNSS MAX-M10S
Size: A4
Software: KiCad E.D.A. 9.0.6

Rev:
Id: 11/16

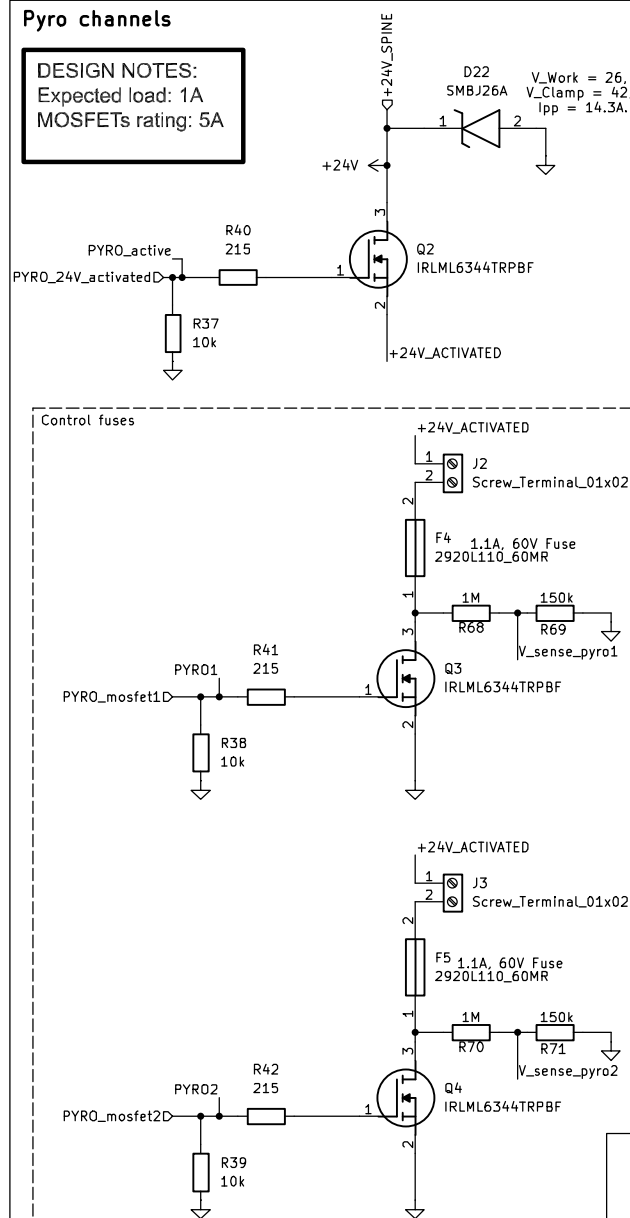




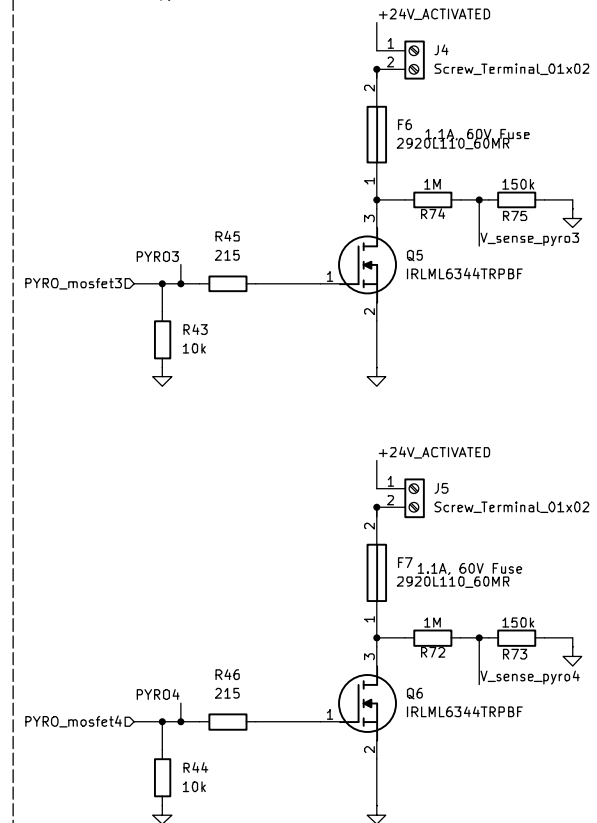
DESIGN NOTES:
 PYRO CONTINUITY - ADC SENSE
 Divider: 1.0 MΩ (top) / 150 kΩ (bottom) → ADC
 Test current (through igniter): ≈ 21 μA @ 24 V (safe, no ignition)
 ADC voltage if igniter present: ≈ 3.1 V
 ADC voltage if igniter open / missing: ≈ 0 V
 Conditions (MOSFET OFF):
 Igniter connected → ADC ≈ 3.1 V → Continuity OK
 Igniter open / fuse blown → ADC ≈ 0 V → No continuity
 RMK :Continuity check valid only when MOSFET is OFF (during firing ADC goes low by design).

Pyro channels

DESIGN NOTES:
 Expected load: 1A
 MOSFETs rating: 5A



Redundant Control pyros



Pyros

Author: Max Wei Ciriani
 Date: 2025-11-23

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Sheet: Pyro

Size: A4

Software: KiCad E.D.A. 9.0.6

Rev:

Id: 12/16



DET_A,B = Card detect switch pins. Behaviour:

- open circuit when no card
- short circuit when card (pulled low)

The GPIO must have an internal resistor (~47k).
GPIO = HIGH = No card
GPIO = LOW = Card

- CMD
- D[0,1,2,3]

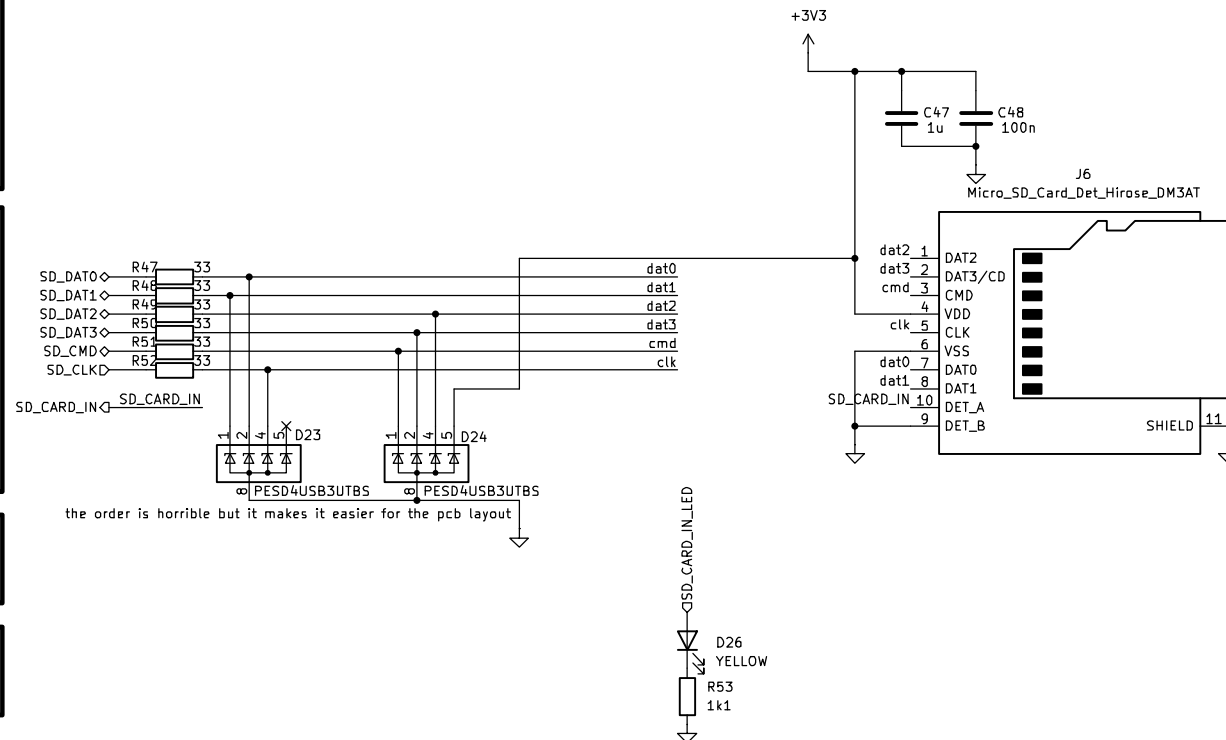
Do not pull up CLK. Leave unpulled in cubelDE

The pins need a pull up resistor of about 10k-100k, the internal pull-ups provide around 40-50k. 47k is a typical value for external pull-ups. -> pullup in cubeIDE

Add 330ohm resistors in serie to every logic line,
So CMD, D[0,1,2,3] and CLK

ESD protection: Low capacitance TVS diodes (<5pf), with working voltage V_RWM around 5V

Try to keep DAT[0,1,2,3], CLK and CMD lines around the same lenght. <10mm difference is recommended for a typical application.

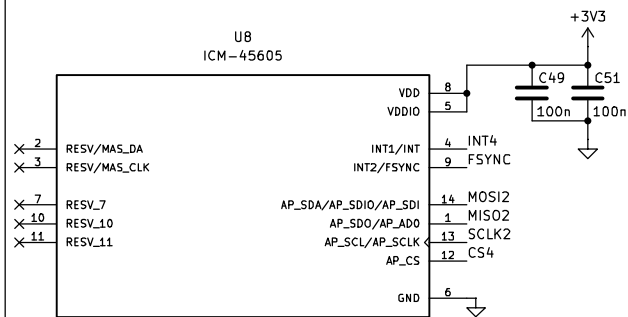


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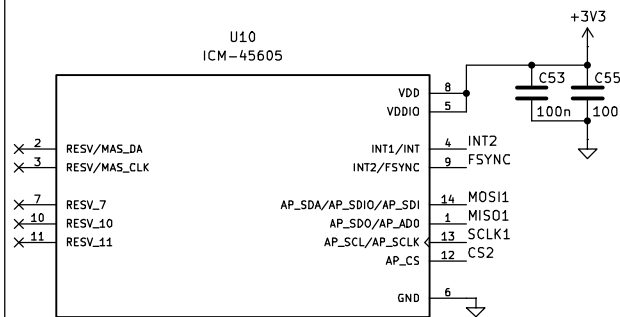
Rev:
Id: 13/16



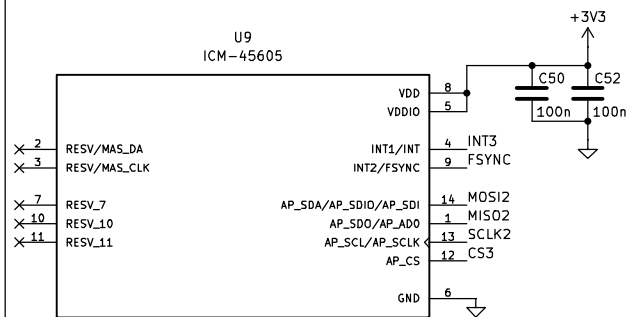
ICM-45605 #4



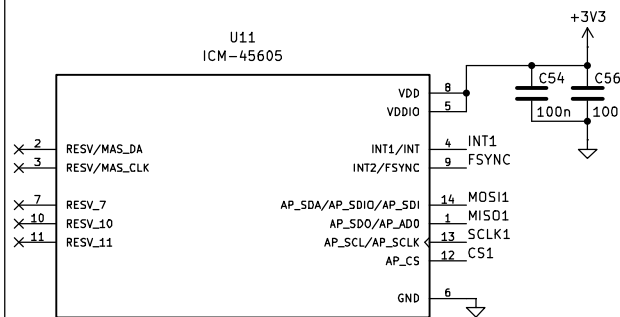
ICM-45605 #2



ICM-45605 #3



ICM-45605 #1



DESIGN NOTES:
100n capas are X7R

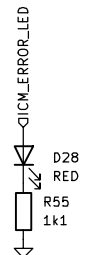
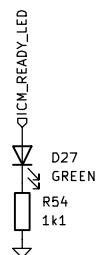
DESIGN NOTES:
All the ICM's share the same
FSYNC line

DESIGN NOTES:
Recap of the SPI protocol:
CLK / SCK : clock
CS : chip select. Goes high
when selected, low otherwise
MOSI : Master to Slave
MISO : Slave to Master

DESIGN NOTES:
The datasheet mentions SDI
and SDO, the equivalents are:
SDI = MOSI
SDO = MISO

DESIGN NOTES:
In case of SPI malfunction, the 4 sensors are
divided in two buses, sensors 1,2 have one
bus, and sensors 3,4 have another bus.

Labels	
MOSI1	ICM_MOSI1
MISO1	ICM_MISO1
SCLK1	ICM_SCLK1
MOSI2	ICM_MOSI2
MISO2	ICM_MISO2
SCLK2	ICM_SCLK2
FSYNC	ICM_FSYNC
CS1	ICM_CS1
INT1	ICM_INT1
CS2	ICM_CS2
INT2	ICM_INT2
CS3	ICM_CS3
INT3	ICM_INT3
CS4	ICM_CS4
INT4	ICM_INT4



ICM-45605

Author: Max Wei Ciriani
Date: 2025-11-23

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Sheet: ICM-45605

Size: A4

Software: KiCad E.D.A. 9.0.6

Rev:

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DESIGN NOTES:

Route D+ and D- as 90 ohms differential pair. And match the length.

DESIGN NOTES:

We are using 2 USB communication lines:
USB OTG FS (Full-speed) = 12Mbps (internal USB PHY in STM32)
USB OTG HS (High-speed) = 480Mbps (needs an external USB PHY)

DFU_USB-A (Bootloader port) → OTG_USB_FS
VCP_USB-A (Virtual COM port) → OTG_USB_HS

We will actually be using normal USB and not USB OTG, because the PC will always be configured to be the host, and the STM32 to be the device. Meaning the PC will push VBUS to the STM32.

DESIGN NOTES:

Internal FS OTG PHY support

External HS or HS OTG operation supporting ULPI in SDR mode (OTG_HS1 only)
The OTG PHY is connected to the microcontroller ULPI port through 12 signals. It can be clocked using the 60 MHz output.

INFO:

Embedded USB_DP pull-up value during idle: $R_{PUI} = 1250$.
Embedded USB_DP pull-up value during reception: $R_{PUR} = 2300$
Output driver impedance: $Z_{DRV} = 36$

INFO:

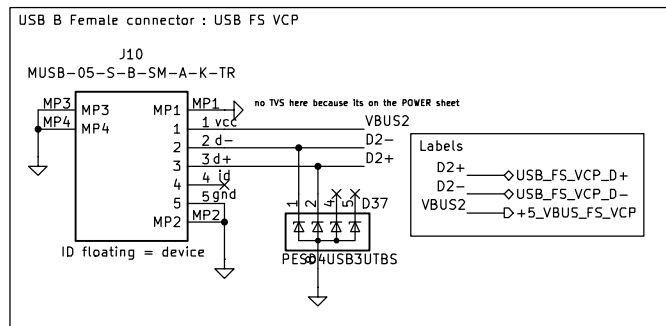
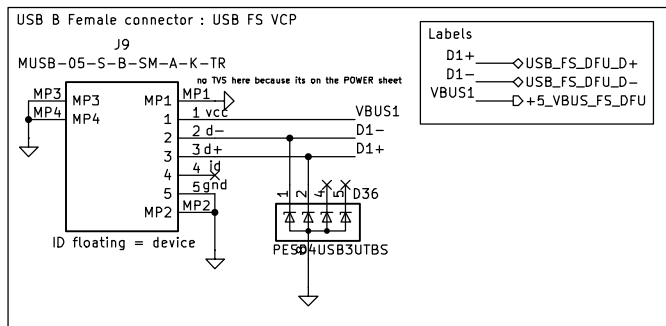
USB OTG adds a fifth pin in its connector, not a fifth wire in the cable.

Both plugs include an ID pin, which is essential for determining the roles of the connected devices, including which will act as the host and which will function as the peripheral. The Standard-A plug has a grounded ID pin. The device it is attached to will be designated as an OTG A-device, which will act as a USB host to supply power and manage the data transfers. The micro-B plug has a floating ID pin. The device attached to this plug will be designated as an OTG B-device, where it will become a USB peripheral and consume power and respond to the host's commands.

DESIGN NOTES:

ULPI Output Clock config:
Square-wave 3V-3.6V reference clock on REFCLK. CLOCK will output a 60Mhz

In the end, we are using 2 USB FS connections, so the notes on the usb hs are not necessary. Nevertheless, I will keep them



USB OTG

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Sheet: USB OTG

Size: A4

Software: KiCad E.D.A. 9.0.6

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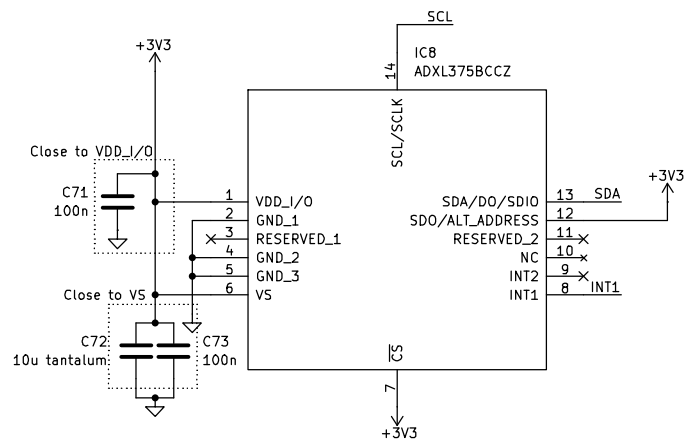
ADXL375

DESIGN NOTES:

Physically place the ADXL close to a mounting point to prevent undampened vibrations

DESIGN NOTES:

CS high = i2c mode
ALT high => i2c address = 0x1D
Write address = 0x3A
Read address = 0x3B



Labels

INT1 -> INT1_ADXL375
ALERT -> TMP_ALERT
SDA -> SDA_tmp_adxl
SCL -> SCL_tmp_adxl

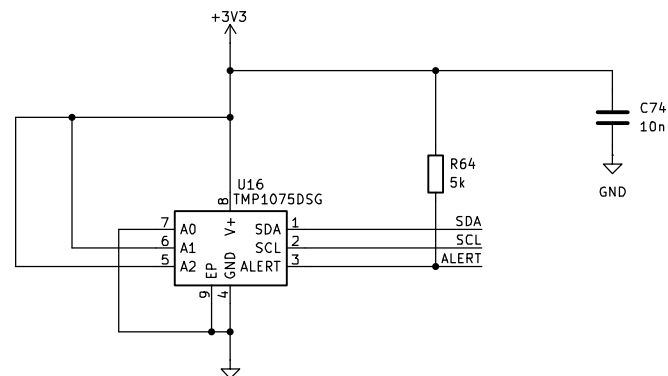
TMP0175N

DESIGN NOTES:

The TMP1075DSG is the TMP1075, not to be confused with the TMP1075N which does not have the A1 and A2 pins.

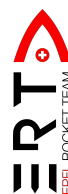
DESIGN NOTES:

Adresse i2c: 0x4f (cf doc page 14)



DESIGN NOTES:

The Alert pin should be configured in comparator mode (TM=0, POL=0): it's a open drain output that will be driven to low (~0.15V) when the temperature exceeds T-HIGH.



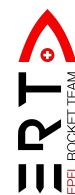
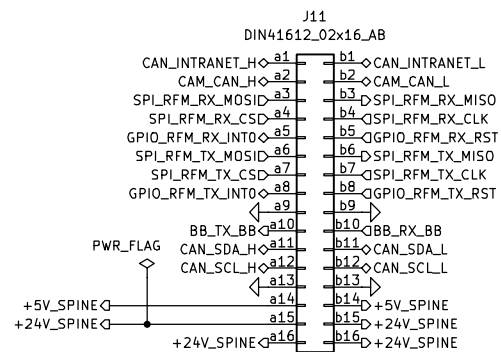
ADXL and TMP

Author: Max Wei Ciriani
Date: 2025-11-23 ©EPFL Rocket Team 2026

Sheet: ADXL375_and_TMP1075N
Size: A4
Software: KiCad E.D.A. 9.0.6

Rev: 15/16



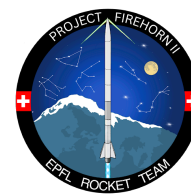


Spine connections

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Sheet: SPINE
Size: A4
Software: KiCad E.D.A. 9.0.6

Rev:
Id: 16/16



NAME	SOURCE	LEVEL
+3V3	LDO	??V ± ??%
+5V	SPINE	??V ± ??%
+5V	USB POWER	??V ± ??%
+24V	BATTERY	??V ± ??%

Power Sequencing Diagram of your PCB here

1. _____

2. _____

3. _____

4. _____

5. _____

6. _____

7. _____

8. _____

9. _____

10. _____

11. _____

12. _____

13. _____

14. _____

15. _____

16. _____

17. _____

18. _____

19. _____

20. _____

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